

Event Classification of accelerometer Data for Industrial Package Monitoring with Embedded Deep Learning

Make industrial packages reusable and autonomous

Single-used packages transport

In Europe, Goods transport by road represented **more than 13 millions of thousands of tonnes in 2023¹**. Most of the transport packages are **single-used packaging**.

Packaging: an important contributor to plastic waste and greenhouse gas emission.



Rethinking industrial packages



Create **reusable** packages.



Example : industrial
reusable package¹

The objective: replace single-use packages to multi-use reusable packages.

➡ Contribute to **industrial decarbonization**

¹By Ranjeetdighe - Own work, CC BY-SA 4.0, <https://commons.wikimedia.org/w/index.php?curid=40095825>

Reusable Packages Challenges

The main challenge for industries : *make it profitable ! But...*



**High Cost of
Reusable Packages**



**Packages can be lost in
transit**



**A relative amount of
packages get stolen**

Tracking packages

Solutions explored :



GPS: tracking accuracy gets bad in indoors or harsh environments.



Time difference of arrival: a dense network of IoTs and antennas is required with a high chance of interference in areas around city center.

Global localization is complicated, so we need something else.

Implying logistic actors

Objective: know which logistic actor is responsible of the package at a given time.

Makes logistic actors more **reliable**.

How does it works ?



1. IoTs are placed under packages.
2. IoTs track who is responsible of packages

Advantages of the solution:

- No use of a GPS
- Last several years
- No human intervention needed

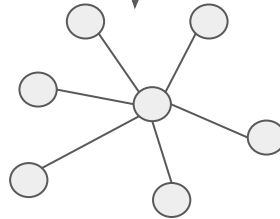
IoT detects a movement

AI model detects environment (on a truck,
on a forklift, etc.)



A new environment= A new logistic actor is responsible of the package

“Are there other IoTs around me ?”



Detects its in a group of IoT

Neighbor graph

Send information to the server

Detecting a new environment

Encountered challenges

Dataset collection



Figure 1: Data loggers placed inside a package at different locations

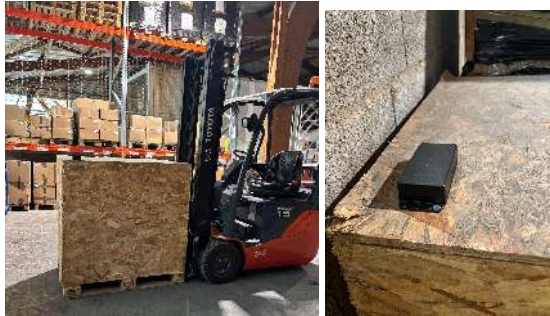
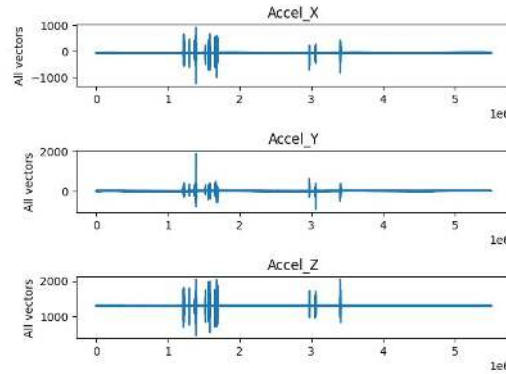
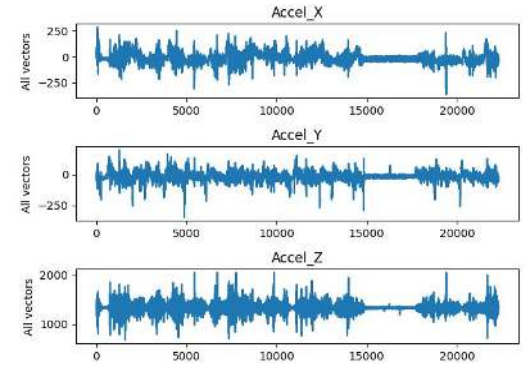


Figure 2: A datalogger is placed on a package carried by a forklift

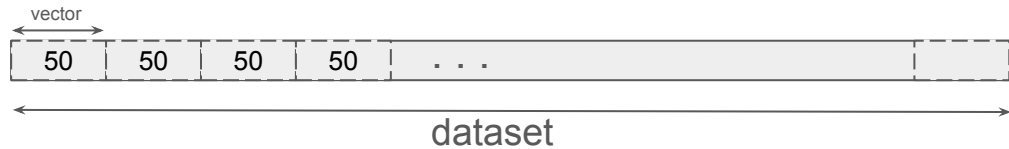


Accelerometer values recorded from a package carried by a forklift



Accelerometer values recorded from a package carried by a truck

Splitting data to keep time coherence :

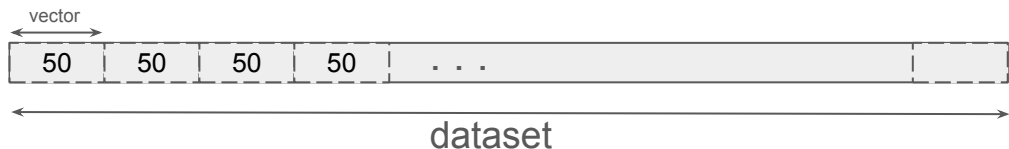


Reducing dataset imbalance

Dataset : accelerometer values

Why is it imbalanced ?

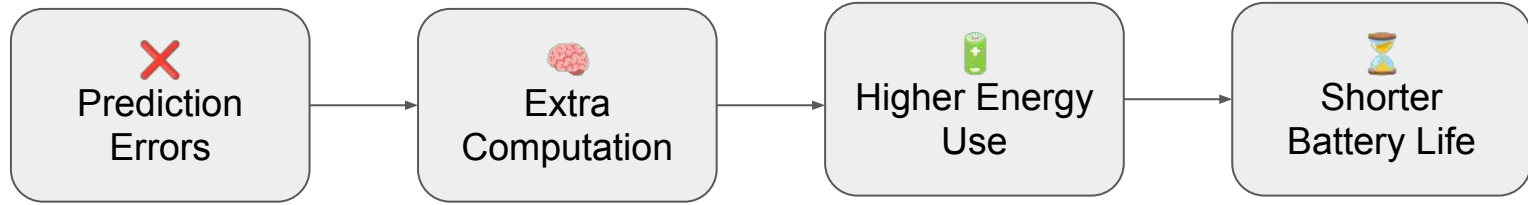
Splitting data to keep time coherence :



Data augmentation technique : ADASYN

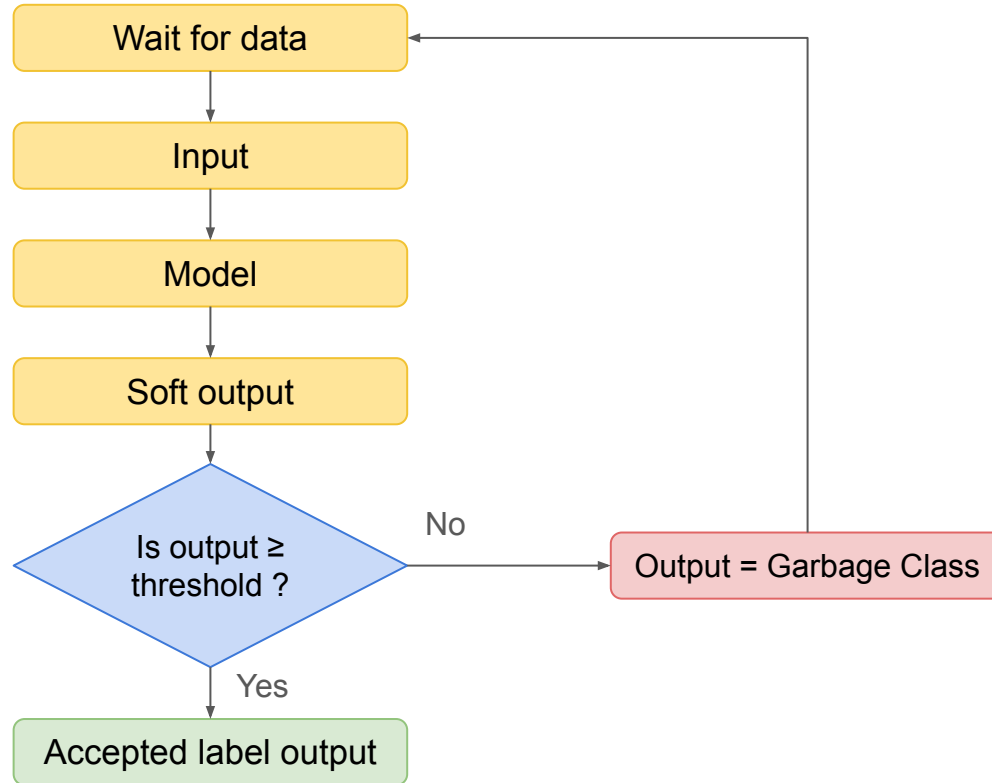
	Forklift		Truck	
	precision	recall	precision	recall
Original dataset	82.17	83.21	91.56	59.98
With ADASYN	98.05	91.17	95.75	96.62

Looking for high precision



Minimizing prediction errors is essential not only for accuracy but also for **energy efficiency** and **long-term device autonomy**.

Apply a Confidence Threshold and Introduce a Garbage Class



Finding the optimal threshold

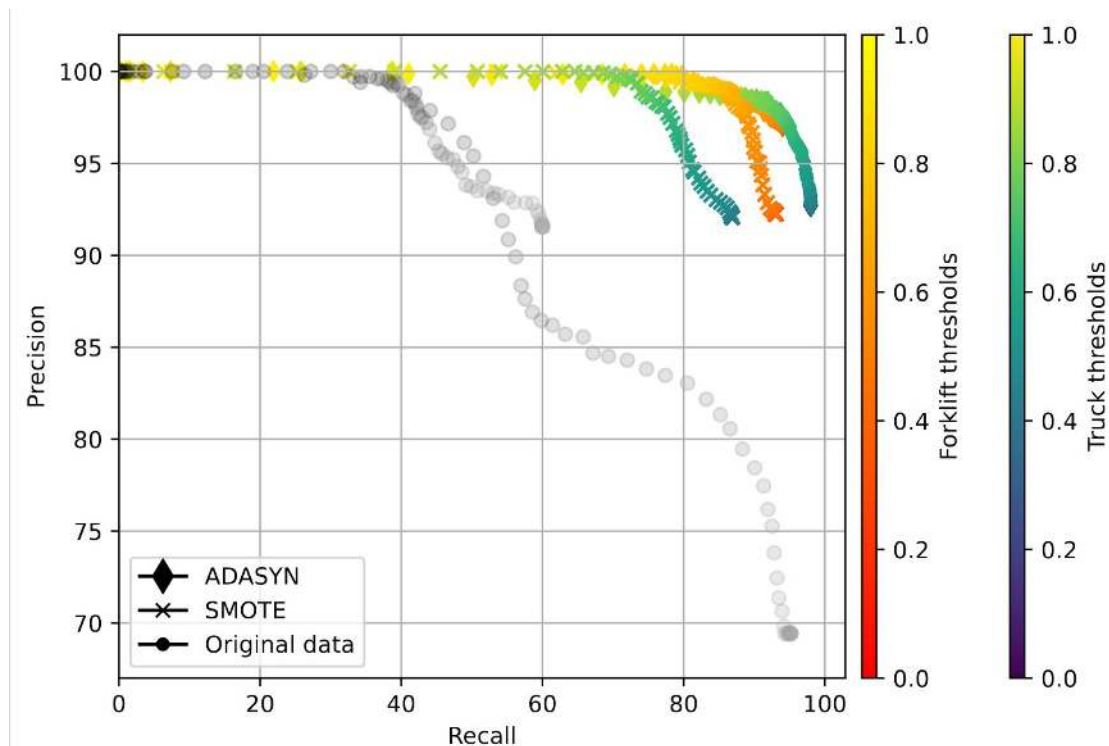


Figure: Precision/recall values with threshold varying, with different DA.

Implementation on embedded system

Compression techniques:

Pruning used : weight rewinding technique

Quantization used : fp32 $\rightarrow$ int8

Specifications:

Microcontroller : ESP32

To deploy the model: LireRT (TFLite)

Configuration	Compressed model size	Forklift		Truck	
		Precision	Recall	Precision	Recall
Original	51 kB	98.05	91.17	95.75	96.62
Pruning	33 kB	93.61	96.12	95.72	93.85
Pruning + Quantization	14 kB	94.54	92.59	95.83	88.66

TABLE II
DEEP COMPRESSION EFFECT ON PRECISION, RECALL, AND MODEL SIZE.

What's next ?

Current version of the system have been tested in march in an industrial environment.

IoT's were placed on packages and could be tracked within the warehouse, and without human intervention.

Results were encouraging. Therefore, the test lead to a future POC next year, on a larger scale (about 6 months).

To improve classification :

- ➔ Implement new sensors and work on multimodal classification
- ➔ Enrich actual environment
 - ◆ Add new labels that the package could encounter
 - ◆ For example : add a Human class

Thank you !

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Paper



Goodflow

Model architecture

